SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I C. Priyanka / 192511074 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: C. Priyanka



Annexure A
Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$p = 17$, $q = 19$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

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2. ElGamal Encryption Analysis

Given:

$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$

Compute the public key

Perform encryption to generate ciphertext pair (C1, C2)

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

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3. RSA Digital Signature Verification

Given:

$p = 11$, $q = 17$, $e = 7$, message $M = 8$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

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4. Diffie-Hellman with Attack Resistance Evaluation

Given:

$p = 29$, $g = 2$

User A private key = 5

User B private key = 12

Compute public keys for both users

Compute the shared secret key

Analyze what happens if an attacker intercepts and replaces public keys

Suggest a cryptographic solution to ensure authenticity

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5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20% 20	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20% 20	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25% 20	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

Accuracy of Calculation	20% 20	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15% 15	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Name : C. priyanka
Reg No : 192511074
Course : CSA 5102
Cryptography

CO_2 AT1

1. RSA Key Generation and Security Analysis

Given,

- P = 17
- 9 = 19
- Public exponent e = 5
- Message M = 7

Step 1 : calculate n

RSA modulus is calculated by multiplying the two prime numbers.

$$n = p \times q$$
$$n = 17 \times 19 = 323$$

Step 2 : Calculate Euler's Totient Function.

The Formula is

$$\phi(n) = (p-1)(q-1)$$

Substituting the values :

$$\phi(n) = (17-1)(19-1)$$
$$= 16 \times 18$$
$$= 288$$

Step 3 : Calculate the Private Key

The private key d Satisfies:

$$ex\ d = 1 \pmod{\phi(n)}$$

Substituting $e = 5$:

$$5d \equiv 1 \pmod{288}$$

Using the extended Euclidean Algorithm, we obtain:

$$d = 173$$

Verification:

$$5 \times 173 = 865 = 3(288) + 1$$

Therefore, the Private Key is Correct

Step 4: Encrypt the Message

The Encryption formula is:

$$C = M^e \bmod n$$

Substituting the values:

$$C = 7^5 \bmod 323$$

$$7^5 = 16807$$

$$16807 \bmod 323 = 11$$

Therefore, the cipher text is 11.

Step 5: Decrypt the Message.

The decryption formula is

$$M = c^d \bmod n$$

Substituting the values:

$$M = 11^{173} \bmod 323$$

using modular exponentiation :

$M = 7$

The decrypted message matches the original Message.

Step 6: Security Analysis

Rsa security depends on the difficulty of factoring a large modulus into its prime factors.

In this example:

$$323 = 17 \times 19$$

Since these factors are small, an attacker can easily find them and calculate the private key.

Security improvements:

1. Use Sufficiently large RSA Keys, typically 2048 bits (or) more.
2. Use secure padding, Such as RSA-OAEP for Encryption.
3. Keep the private Key Secret
4. Use trusted cryptographic libraries.
5. protect private Keys against theft and unauthorized access.

Conclusion:

Rsa Provides a method for Public-Key encryption and Private-Key decryption. The given message is encrypted into ciphertext it and Successfully decrypted back to it. However, the small prime numbers make this example unsuitable for real-world Security.

2. Diffie Hellman key Exchange and Attack Problem

Given,

- prime Numbers : $p = 29$
- Generator : $g = 2$
- User A's Private key : $a = 5$
- User B's Private key $b = 12$

Step 1 : Calculate User A's Public key

$$A = g^a \bmod p$$

$$A = 2^5 \bmod 29$$

$$A = 32 \bmod 29$$

$$\boxed{A = 3}$$

Therefore, User A's public Key is 3.

Step 2 : calculate User B's Public Key

The formula is

$$B = g^b \bmod p.$$

$$B = 2^{12} \bmod 29$$

$$B = 4096 \bmod 29$$

$$\boxed{B = 7}$$

Therefore, User B's Public Key is 7.

Step 3: Exchange Public Keys

User A sends public key 3 to User B.

User B sends public key 7 to User A.

The private keys a = 5 and b = 12 are never transmitted.

Step 4: Calculate User A's shared secret

$$K_A = B^a \bmod p$$

$$K_A = 7^5 \bmod 29$$

$$7^5 = 16807$$

$$16807 \bmod 29 = 16$$

Therefore, User A's shared secret is 16.

Step 5: calculate User B's shared secret

$$K_B = A^b \bmod p$$

$$K_B = 3^{12} \bmod 29$$

Using Modular exponentiation:

$$3^4 \bmod 29 = 23$$

$$3^8 \bmod 29 = 27$$

Therefore:

$$3^{12} \bmod 29 = (3^8 \times 3^4) \bmod 29$$

$$= (7 \times 23) \bmod 29$$

$$K_B = 16$$

Step 6: Verify the shared secret

User A calculates:

$$K_A = 16$$

User B calculates:

$$K_B = 16$$

Hence:

$$\boxed{K_A = K_B = 16}$$

Both users successfully generate the same sh
secret.

Conclusion:

Diffie-Hellman allows two users to establish a common secret key to over a public channel without transmitting their private keys. In this Pro example, both users obtain the shared secret 16. Authentication is essential to protect against Man-in-the Middle attacks.